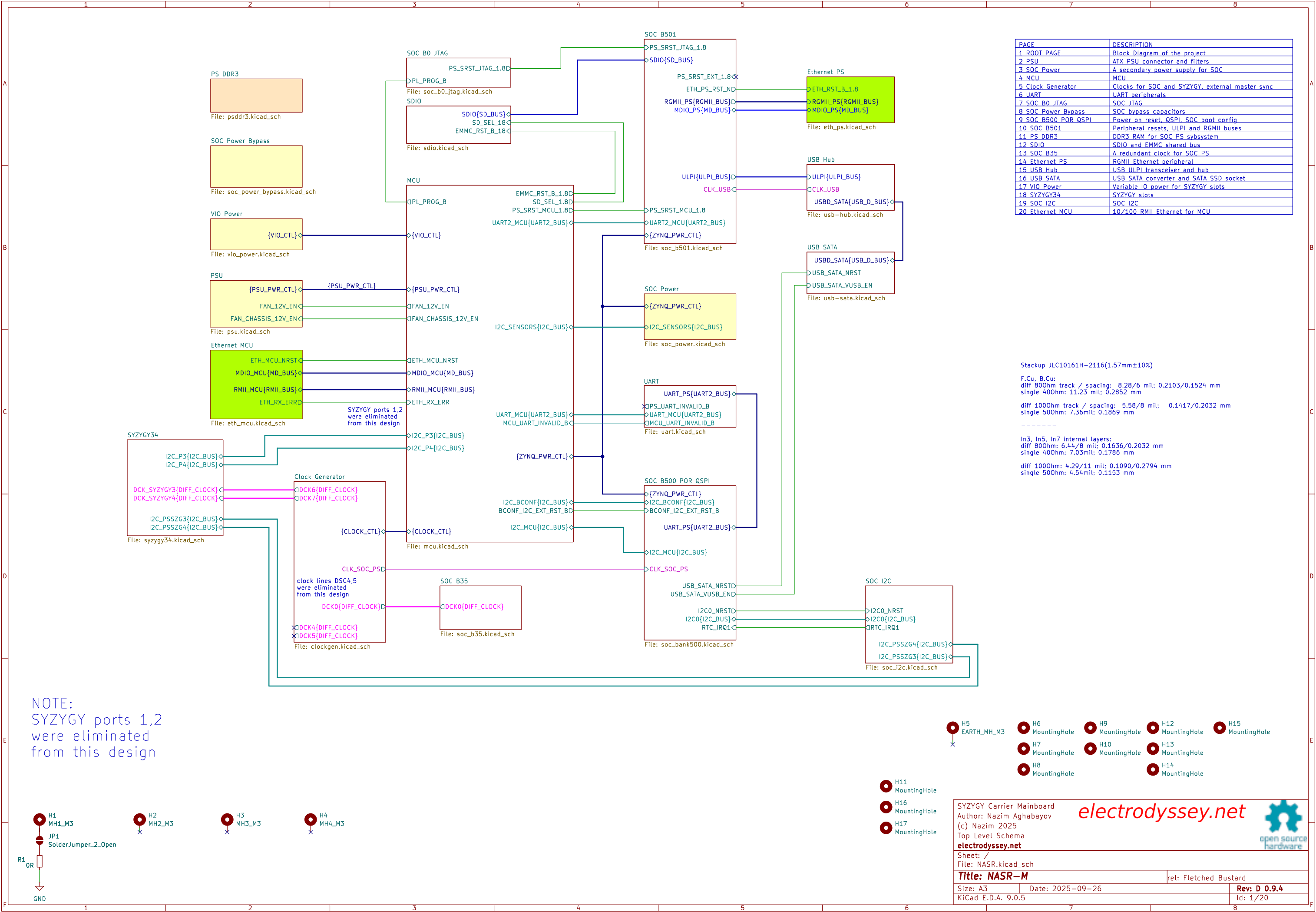
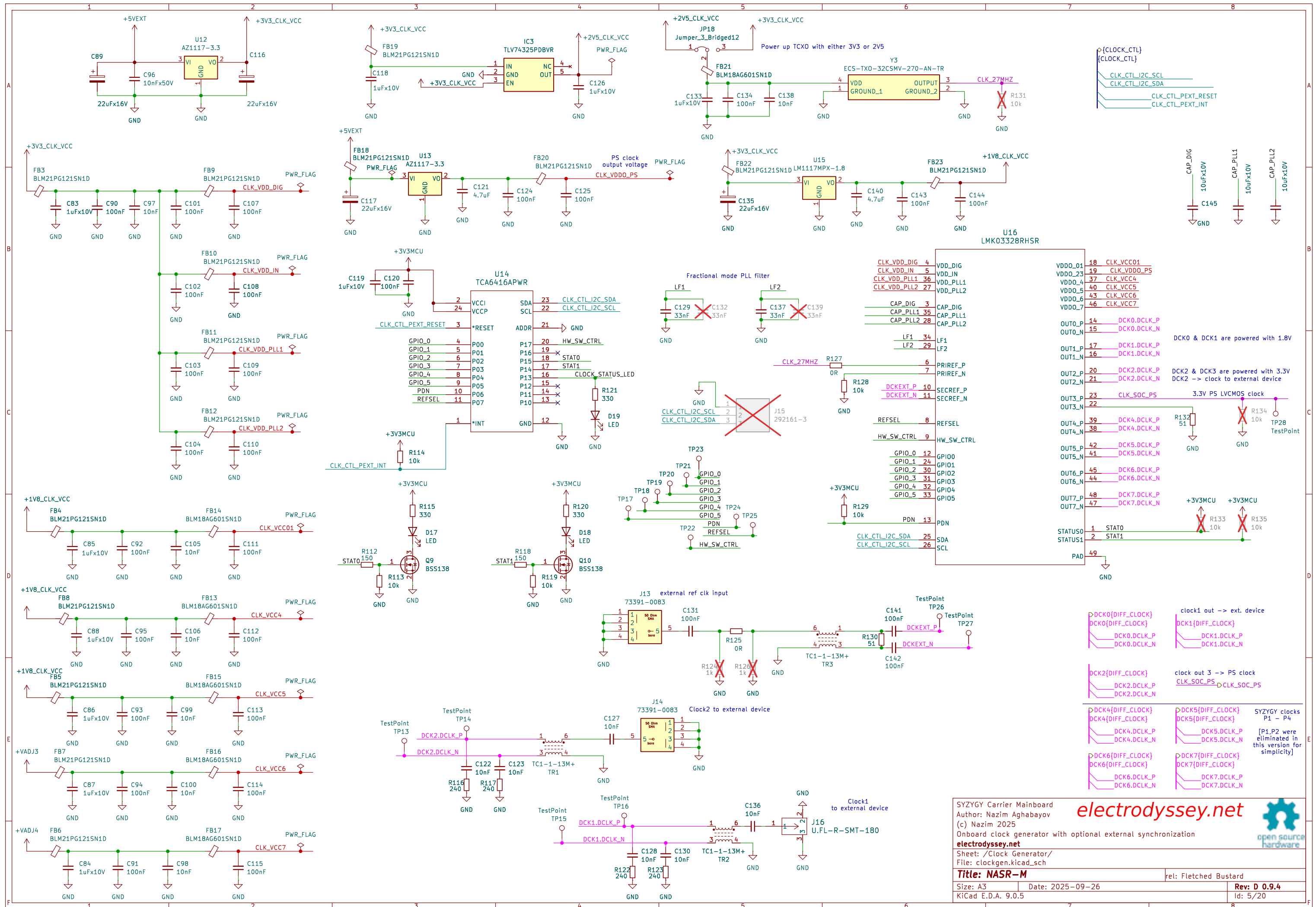
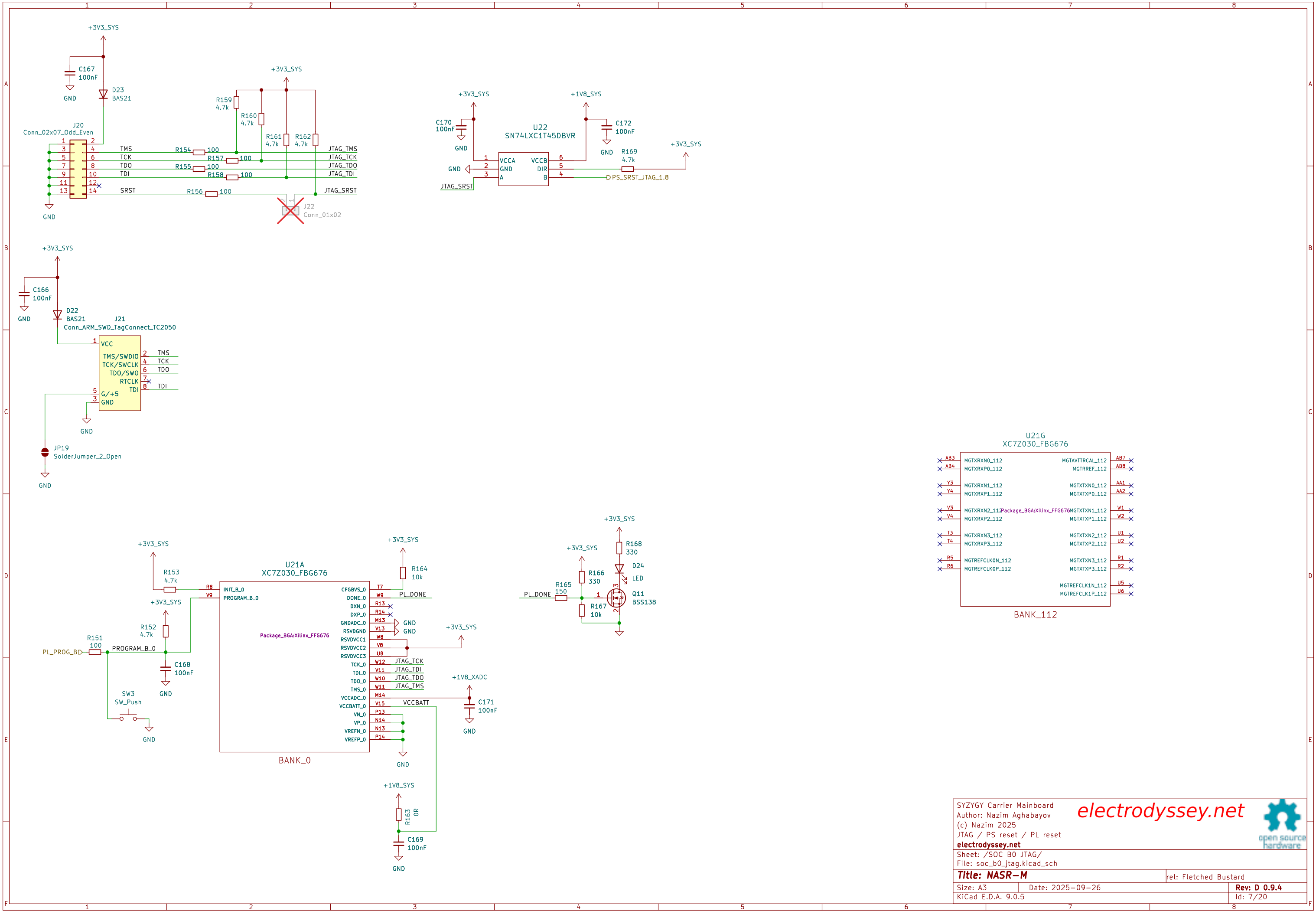
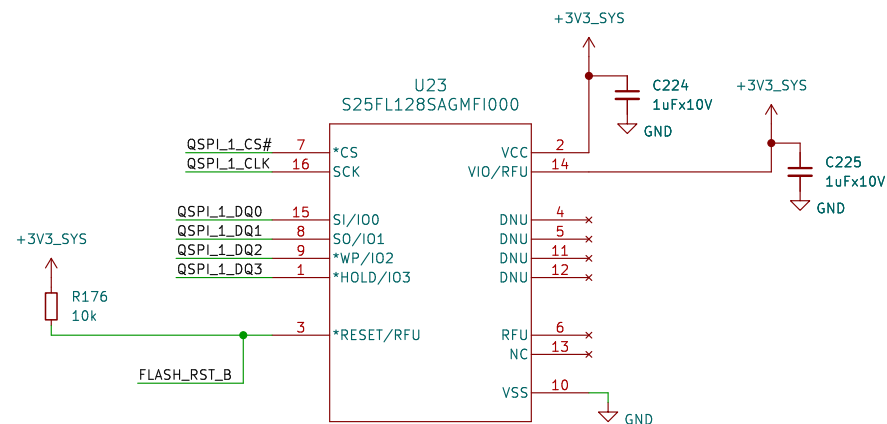
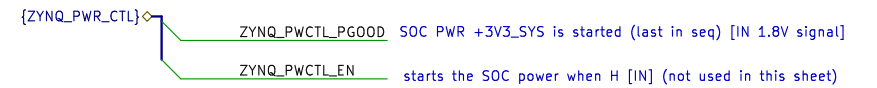


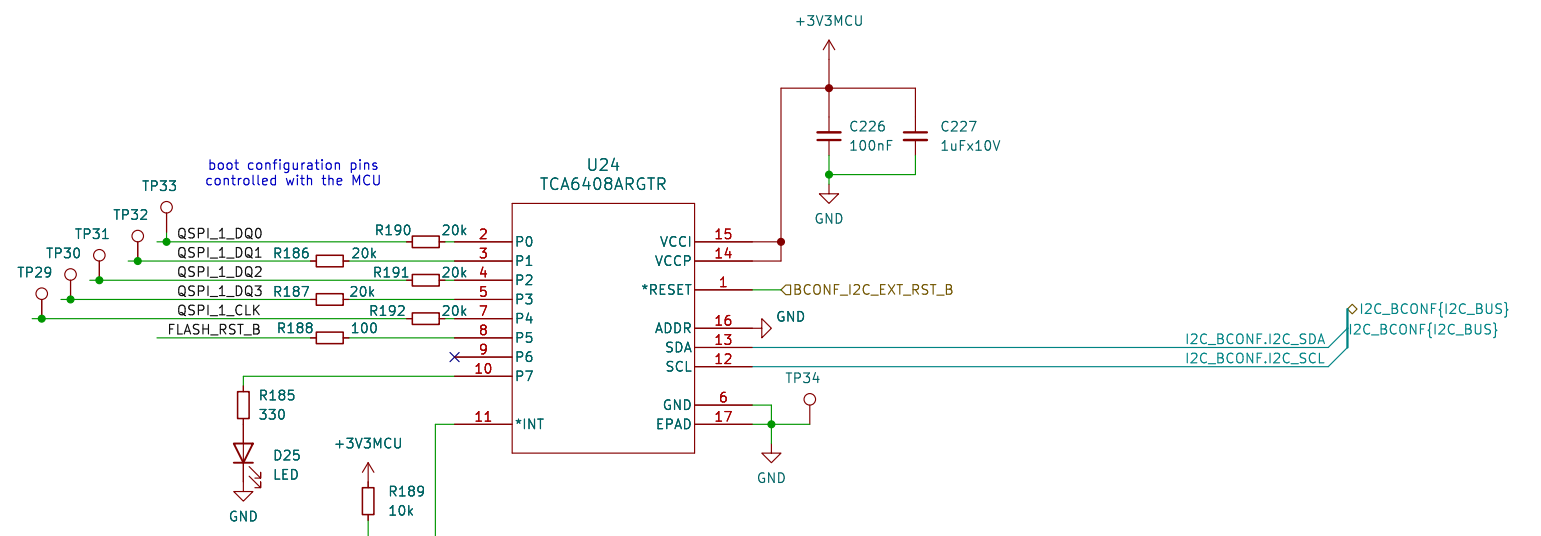


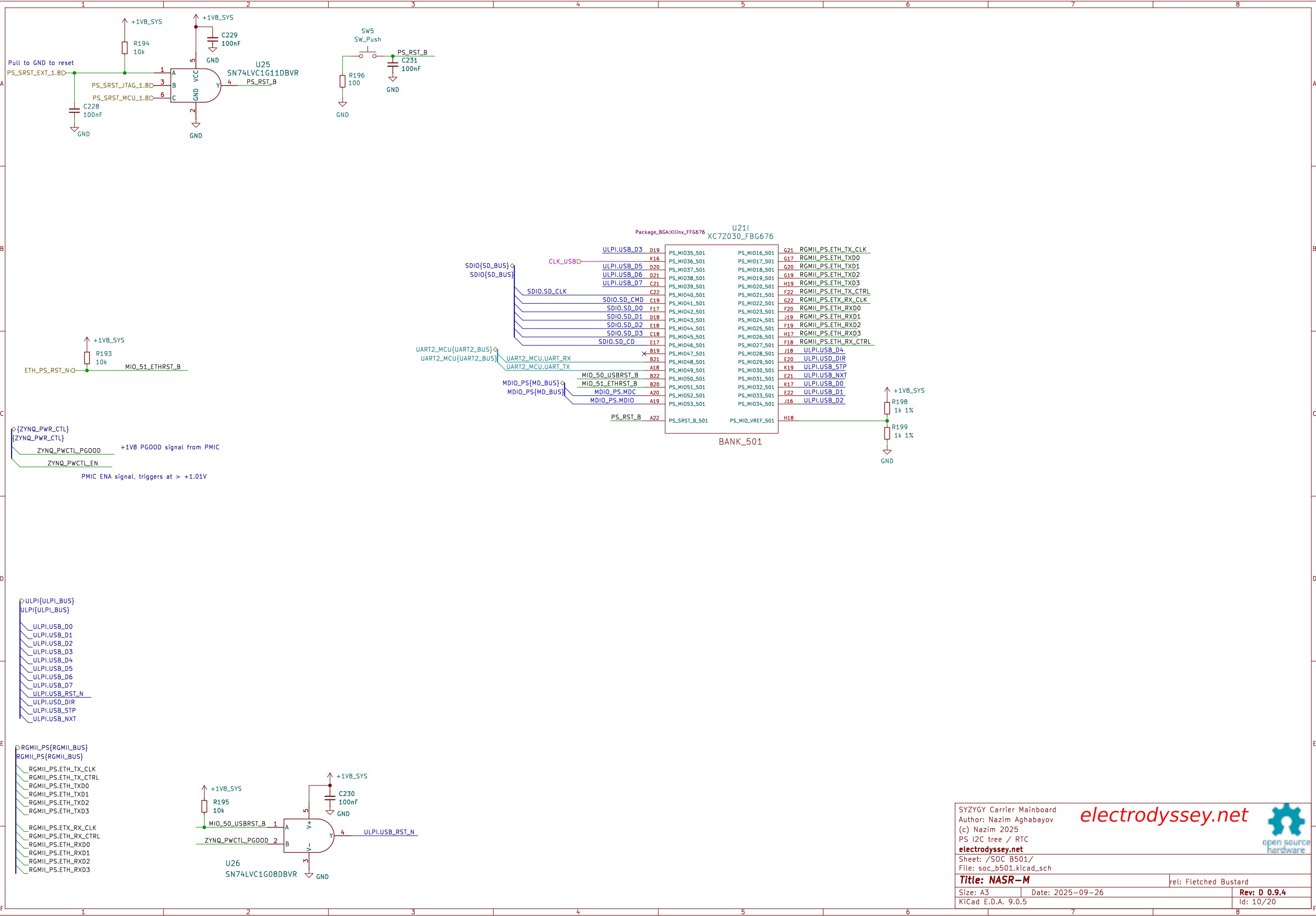


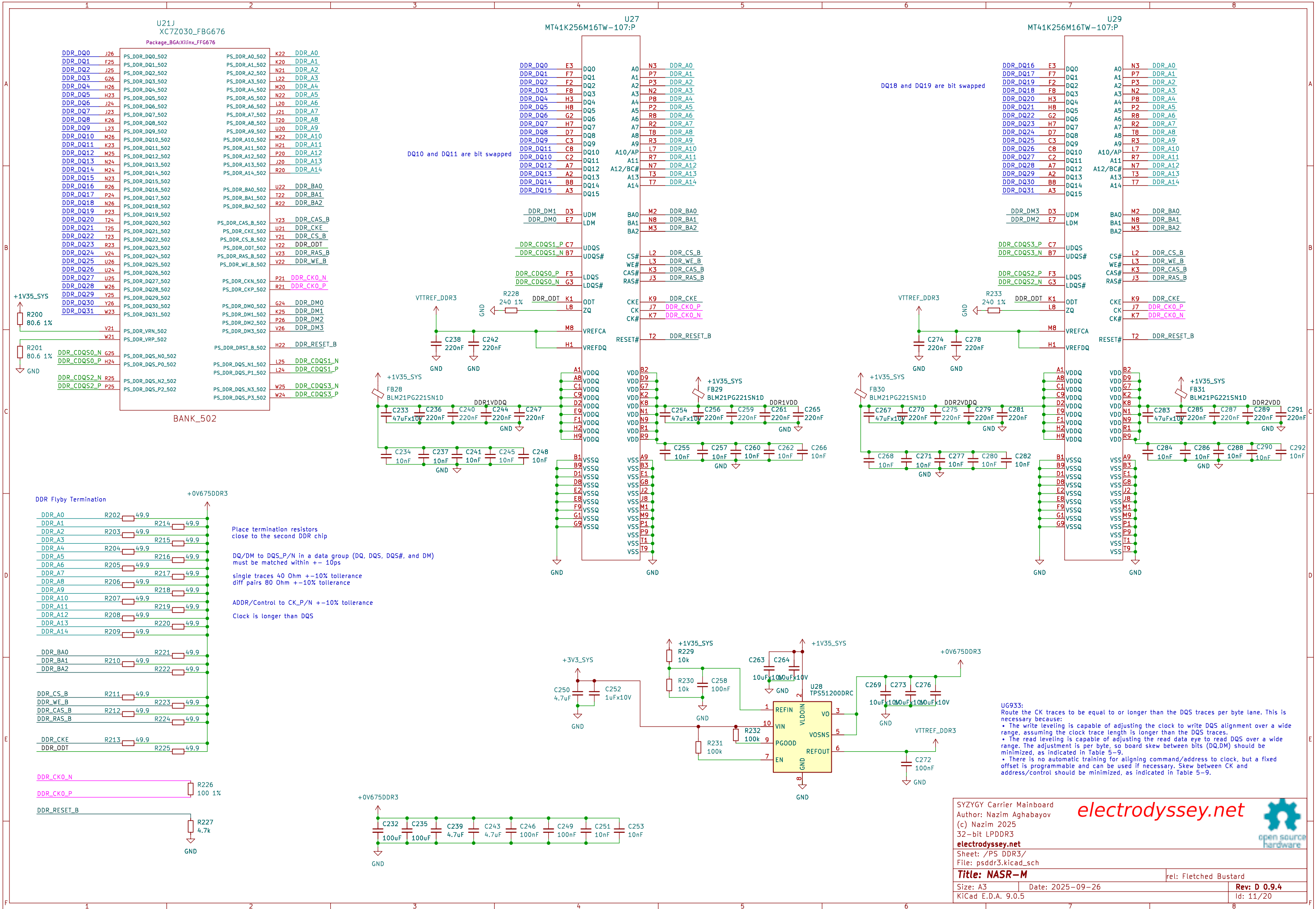


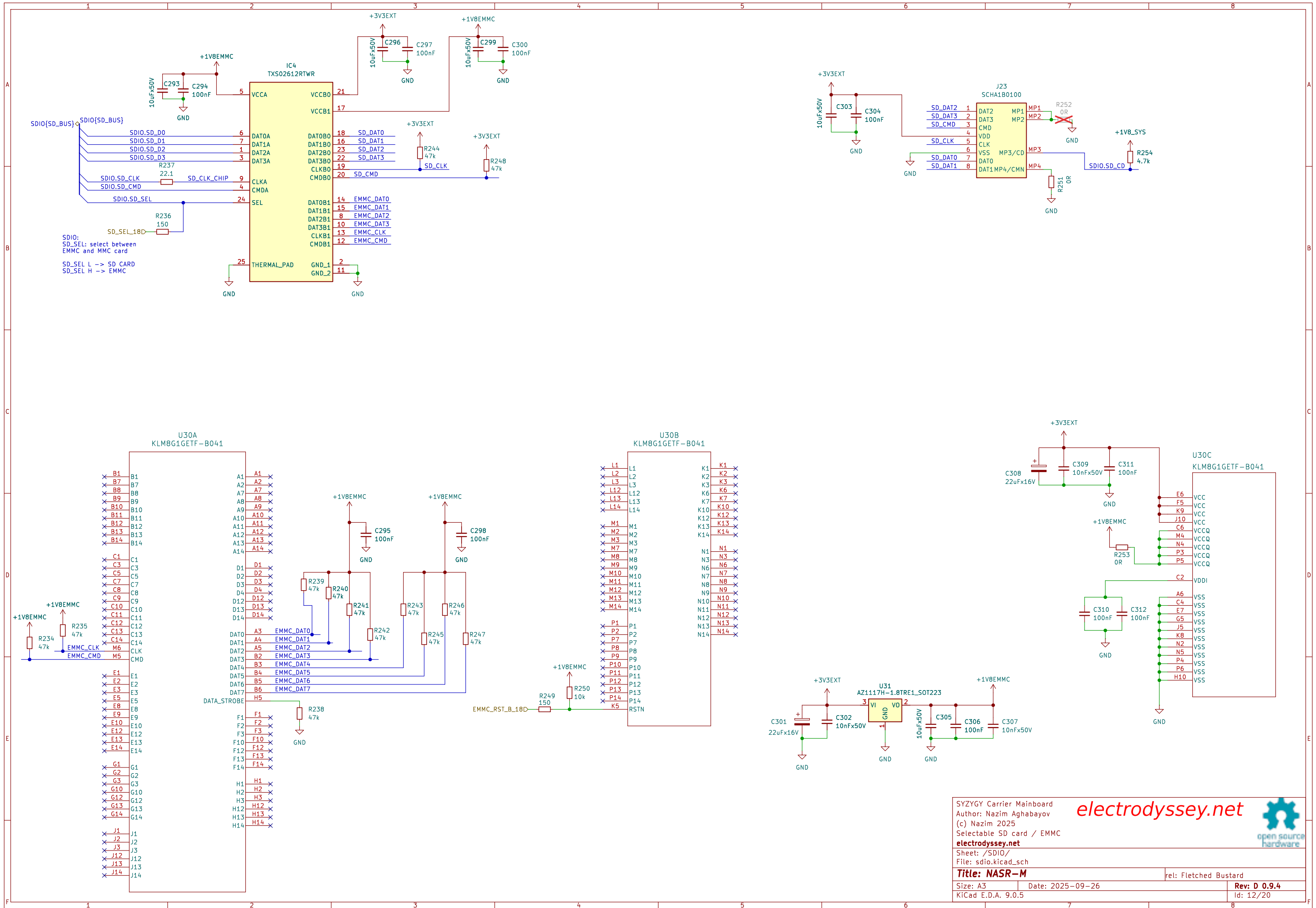


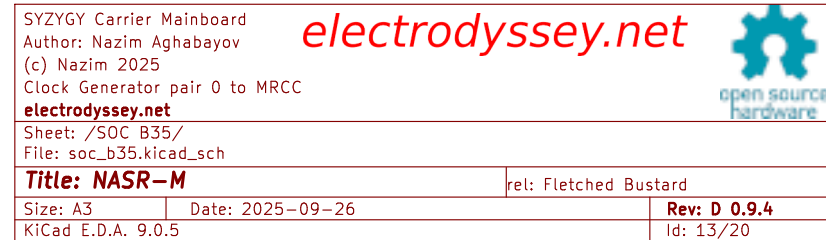
Pin-signal / Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[0]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[3]
Boot Devices							
JTAG Boot Mode; cascaded is most common ⁽¹⁾			0	0	0		JTAG Chain Routing ⁽²⁾ 0: Cascade mode 1: Independent mode
NOR Boot ⁽³⁾			0	0	1		
NAND			0	1	0		
Quad-SPI ⁽³⁾			1	0	0		
SD Card			1	1	0		
Mode for all 3 PLLs							
PLL Enabled		0	Hardware waits for PLL to lock, then executes BootROM.				
PLL Bypassed		1	Allows for a wide P5_CLK frequency range.				
MIO Bank Voltage⁽⁴⁾							
	Bank 1	Bank 0	Voltage Bank 0 includes MIO pins 0 thru 15.				
2.5 V, 3.3 V	0	0	Voltage Bank 1 includes MIO pins 16 thru 53.				
1.8 V	1	1					

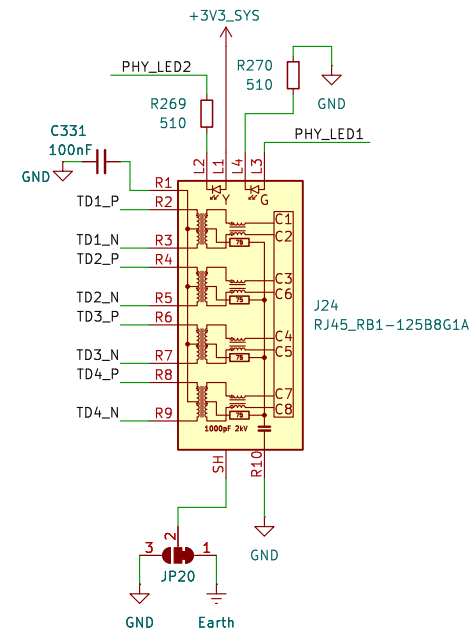
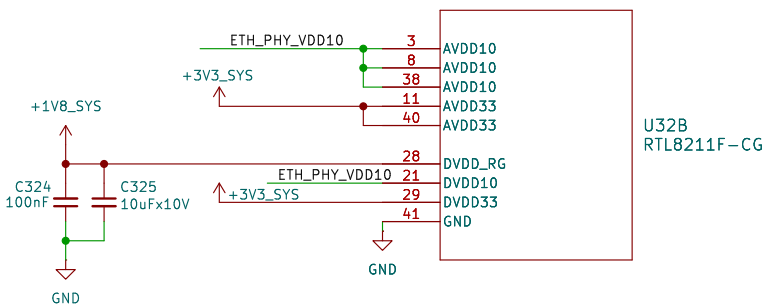
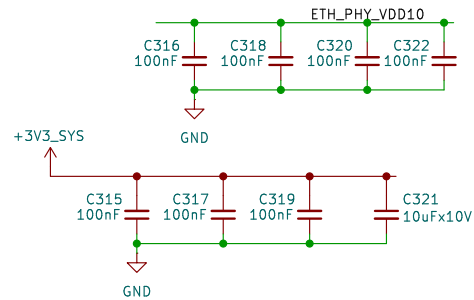
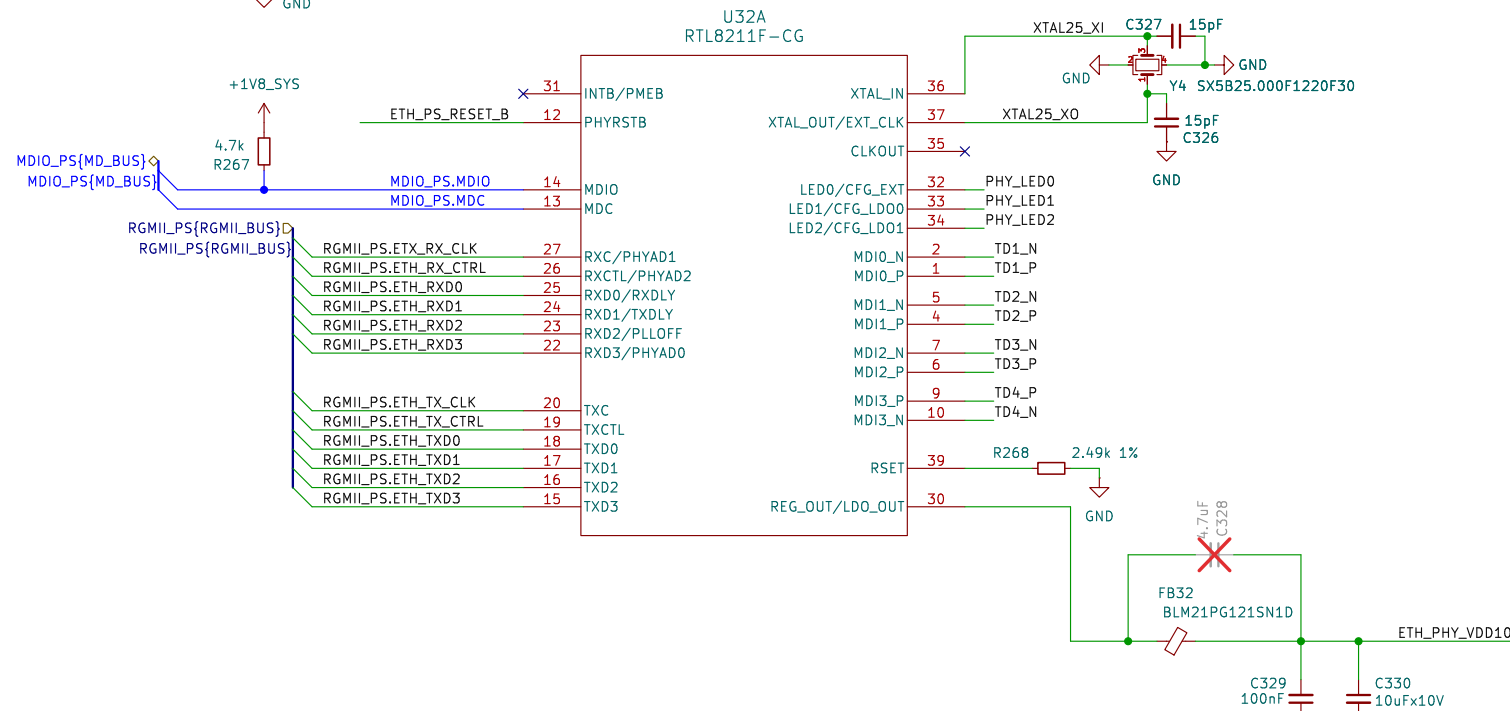
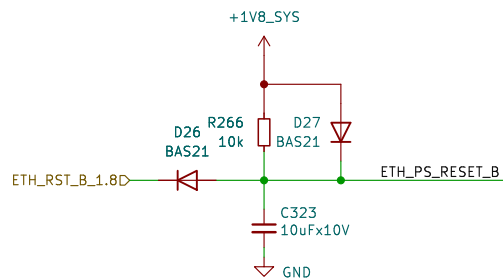
[illegible]











Configuration straps:

LED0 CFG_EXT
LED1 CFG_LDO0
LED2 CFG_LDO1

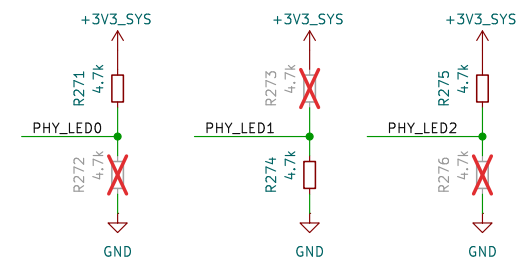
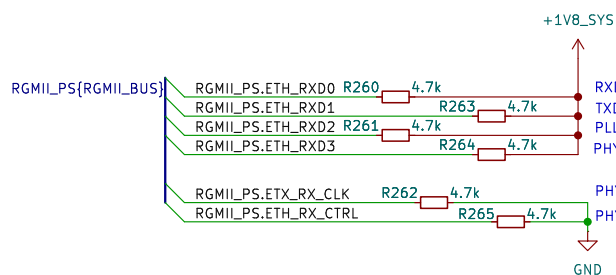


Table 10. CONFIG Pins vs. Configuration Register

CONFIG Pin	Configuration
RXD3	PHYAD[0]
RXC	PHYAD[1]
RXCTL	PHYAD[2]
RXD2	PLLOFF
RXD1	TXDLY
RXD0	RXDLY
LED0	CFG_EXT
LED1	CFG_LDO[0]
LED2	CFG_LDO[1]

Table 11. Configuration Register Definitions

Configuration	Description
PHYAD[2:0]	PHY Address. PHYAD sets the PHY address for the device. The RTL8211F(1)/RTL8211FD(1) supports PHY addresses from 00001 to 00111. <i>Note 1: An MDIO command with PHY address=0 is a broadcast from the MAC; each PHY device should respond. This function can be disabled by setting Reg24.13=0 (See Table 37).</i> <i>Note 2: The RTL8211F(1)/RTL8211FD(1) with PHYAD[2:0]=000 can automatically remember the first non-zero PHY address. This function can be enabled by setting Reg24.6=1 (See Table 37).</i>
PLLOFF	ALDPS Mode PLL Off Configuration. 1: Stop PLL when entering ALDPS mode (via 4.7k-ohm to DVDD_RG) 0: PLL continue toggling when entering ALDPS mode (via 4.7k-ohm to GND)
TXDLY	RGMII Transmit Clock Timing Control. 1: Add 2ns delay to RXC for RXD latching (via 4.7k-ohm to DVDD_RG) 0: No delay (via 4.7k-ohm to GND)
RXDLY	RGMII Receive Clock Timing Control. 1: Add 2ns delay to RXC for RXD latching (via 4.7k-ohm to DVDD_RG) 0: No delay (via 4.7k-ohm to GND)
CFG_EXT	I/O Pad External Power Source Mode Configuration. 1: Use the external power source for the IO pad (via 4.7k-ohm to 3.3V) 0: Use the integrated LDO to transform the desired voltage for the IO pad (via 4.7k-ohm to GND)



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Sheet: /Ethernet_PS/
File: eth_ps.kicad_sch

Title: NASR-M

rel: Fletched Bustard

Size: A3

Date: 2025-09-26

Rev: D 0.9.4

KiCad E.D.A. 9.0.5

Id: 14/20

